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RESEARCH

Food Equality: Multiproxy Biomolecular Dietary Analysis Shows Unstratified Foodways Among Protohistoric E Baltic Communities

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Abstract
Biomolecular archaeology has significantly advanced our understanding of ancient foodways. However, the inevitable methodological biases of those novel methods might give deviating, even contradictory results. This paper highlights the major challenges of, but also potential solutions for, combining and comparing multiproxy dietary data. A rich dataset covering human skeletal remains, pottery, and zooarchaeological records from habitation and burial contexts from the Late Iron Age (650–1300 CE, LIA) Pada archaeological complex, NE Estonia, was subject to stable isotope, organic residue, and microfossil analysis, further complemented by local zooarchaeological and archaeobotanical records. All those dietary records were combined and compared through statistical modelling approaches to (a) exemplify the limitations and biases of different analytical approaches and (b) build a more cohesive and nuanced understanding of local foodways and their social variations. Our results show that LIA Eastern Baltic communities relied heavily on terrestrial, particularly plant based C3 diets with relatively low proportions of animal-based foods, although animal products were preferred in burial ritual context. Interestingly, no dietary variations were detected across different age and sex categories, proving equal access to main food sources for all community members in Pada. The latter contradicts with previously reported datasets from the same region and emphasises the utmost importance of multilevel dietary analysis not only in terms of dietary proxies, but also for different consumption contexts.

Keywords Biomolecular archaeology · Diet · Stable isotope analysis · Microfossil analysis · Organic residue analysis · Eastern Baltic · Late Iron Age · Early Medieval Period

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Introduction

Biomolecular archaeology has become an intrinsic part of ancient dietary reconstructions. The major advantage of biomolecular dietary analysis is its capacity to reveal the direct dietary input of past populations through single individuals and/or well-refined contexts. The field is vastly driven by bone collagen stable isotope analysis (SIA; Lee-Thorp, 2008; Makarewicz & Sealy, 2015), which has been recently complemented with compound specific (*i.e.* amino acid) stable isotope analysis (CSIA; Evershed *et al.*, 2008; Naito *et al.*, 2016; Webb *et al.*, 2015, 2018), and lipid/organic residue analysis (ORA) of ancient pottery (Cramp *et al.*, 2023; Evershed, 2008a, 2008b). These biomolecular approaches can be further supported by microfossil studies, which target microscopic remains of plants or other ecofacts from either dental calculus (Cristiani *et al.*, 2018; Salazar-García *et al.*, 2021) or pottery foodcrusts (visible burnt food residues on the surface of the vessel; Chen *et al.*, 2023; Lucquin *et al.*, 2023; Teetaert *et al.*, 2024).

Multianalytical studies of ancient diets have proven to be particularly advantageous, because each biomolecular technique by itself is inevitably limited by certain analytical or taphonomical biases. Combining different sample types and analytical approaches can reveal a more cohesive and nuanced picture of ancient foodways. However, the majority of those multianalytical strategies of ancient dietary reconstructions centre around the same dietary proxy—be it skeletal remains (*e.g.* bones SIA and dental calculus analysis) or pottery (*e.g.* analysis of absorbed and visible food residues). Synergetic combination and comparison of different dietary proxies and analytes like SIA, ORA and microfossil records, further complemented by zooarchaeological and archaeobotanical datasets, occur less often (Cramp *et al.*, 2014; McClatchie *et al.*, 2022; Oras *et al.*, 2018, 2023). However, such combinatory approaches using multilevel evidence are particularly relevant for analysing dietary variations in complex subsistence economies relying on both wild and domesticated species, plants and animals. More detailed dietary records are also necessary for tracking socially embedded dietary availabilities and restrictions, including, *e.g.* the subtle nuances of foodways in different consumption contexts or across different social categories.

This paper presents the major challenges and benefits of combined multiproxy and multimethod dietary reconstructions from different archaeological source materials. We build on a rich dataset of skeletal remains and pottery from contemporaneous cemetery and habitation contexts of Pada archaeological complex (NE Estonia) dating to the Late Iron Age (LIA), covering Pre-Viking (550–800 cal CE), Viking (800–1050 cal CE), and Final Iron Age (1050–1200/1300 cal CE; FIA). We aim to exemplify how the combination and comparison of different biomolecular and microfossil records, complemented by zooarchaeological and archaeobotanical information, allow to create a more cohesive understanding of dietary practices on a specific community level. Special methodological emphasis is put on the integration of the results from different dietary proxies and analytical techniques, also through more advanced statistical and modelling approaches. The study material covers inhumation burials from the cemetery, pottery and

animal records from habitation contexts, and pottery concealed as individual grave goods in funerary rituals. The latter provide a unique opportunity to illustrate the possibilities of revealing more refined and socially embedded dietary practices, such as lifetime dietary habits as opposed to burial ritual dietary variations, or differentiation of foodways across different social categories. All in all, the paper serves as a conceptual framework for combined multiproxy and multi-analytical dietary analysis, allowing for a more cohesive, nuanced, and complex understanding of past foodways.

Materials

Pada archaeological complex is a unique set of Late Iron Age sites located in NE Estonia dating to the sixth/seventh to thirteenth centuries CE (Fig. 1: A). The complex was excavated in 1977–1989 as part of the bigger fieldwork campaign led by Toomas Tamla (Tamla, 2008a, 2008b). It contains two hillforts, a settlement site and an inhumation cemetery. The two hillforts comprise of so-called hillfort I dating to the eleventh to thirteenth centuries CE, and hillfort II dating to the sixth/seventh to eleventh century CE. The settlement dates to the seventh to eleventh century CE., *i.e.* it is contemporaneous with the hillfort II. The settlement surrounds hillfort I from SE, E and N, but also extends to the hillfort I area, forming an earlier occupation layer there, *i.e*. FIA hillfort (hillfort I) was built on top of (Pre-)Viking Age settlement.

The Pada inhumation cemetery is dated to the twelfth to thirteenth century CE based on grave goods. A total of 172 graves were excavated in 1986–1989 (Tamla,

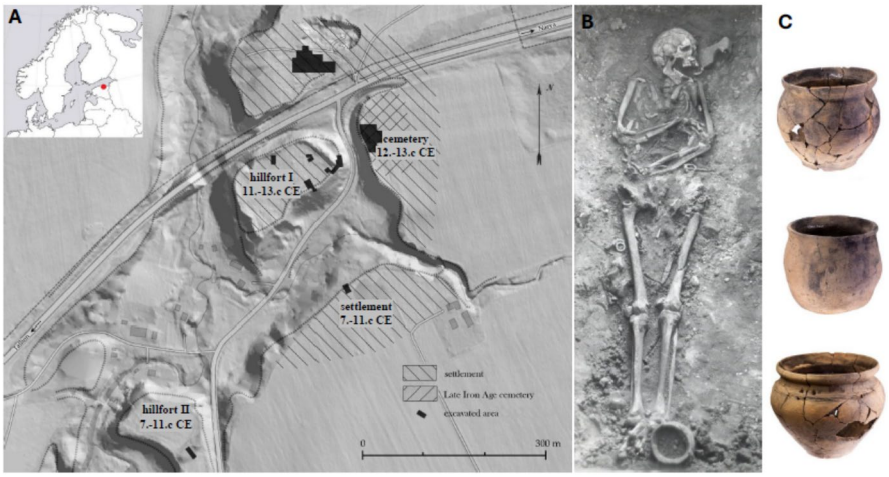

Fig. 1 Location of Pada archaeological complex with **A** map of Late Iron Age sites (from Tvauri, 2012 fig. 20 by Kristel Roog, modified by K. Johanson, E. Oras), **B** example of an inhumation burial XIV with ceramic container from 1987 excavations (Tamla, 2011a), and **C** examples of pottery from Pada inhumation cemetery (AI 5366: XII, XLVI, XLI; photos by Jaana Ratas)

1998, 2011a, 2011b, 2011c). The osteological analysis (including the age and sex determination used in this study) was conducted by Leiu Heapost (Heapost, 2002) using the conventional methods used in Soviet anthropological scholarship in the 1980s, often reported without detailed description of the methods applied. The burials contain both males and females from different age groups. The majority of the burials were accompanied by variable grave goods, some of them richly furnished, including weapons, tools, and jewellery. More importantly for the current study, at least 51 burials had a ceramic container concealed as grave goods with them (Fig. 1: B). Most vessels were placed at the feet (both lower and upper leg regions), although sometimes also by the head of the deceased. The pottery is of typical LIA material containing both wheel-thrown and handmade vessels (Tvauri, 2005), some preserved intact (Fig. 1: C).

A previous study of dietary related material from Pada complex contains the analysis of zooarchaeological records (Maldre, 2007) and a multiproxy analysis of pottery foodcrust from the (Pre-)Viking Age settlement (Chen *et al*., 2023). According to Maldre (2007), the animal remains from settlement contexts excavated in 1980–1982 are dominated by domestic livestock, mostly cattle, sheep, and goats, with horses and pigs being represented to a lesser extent. Wild animals form a small fraction (ca. 3%), of which the majority comes from 36 seal bone fragments. Relying on slaughtering age, the cattle was exploited for dairy and as work animals, but their meat was consumed as well. Fish and bird representation in zooarchaeological records is very limited—only six fish bones and 35 bird bones were reported. This is most likely an archaeological fieldwork bias, as soil sample collection and floatation were not practiced during the excavations. Probably due to the same reasons, no macroscopic plant remains are known from the Pada complex. However, it is evident from contemporaneous LIA sites elsewhere in N Estonia that crop cultivation was well known to these communities with major C3 crops like *Hordeum vulgare* (barley), *Triticum aestivum* (bread wheat), *Secale cereale* (rye), and *Pisum sativum* (pea) being widely represented in archaeobotanical records (Tvauri & Vanhanen, 2016). The consumption of plants can be also presumed by the findings of organic residue and microfossil analysis from (Pre-)Viking Age settlement pottery, which have shown considerable starchy plant evidence—plant-related lipid signals and/or microfossils were detected in the majority of the 24 vessels analysed by Chen *et al*. (2023).

The selection of material for this study (Table 1, and cf. SI Table 1 for details) relied on several considerations. First, we wanted to have some basic information about the major food substances from LIA habitation context, be it in the form of pottery (analysed with ORA) or animal remains, to have some comparative material for the results obtained from the cemetery context. As there is no FIA settlement known from Pada, we could only rely on hillfort materials. Pottery from hillforts, especially hillfort I being contemporaneous with the inhumation cemetery, was therefore analysed to indicate the availabilities of food substances in daily contexts. Animal remains from the Pada hillfort are very scarce and have not been studied in detail. Hence, we could only rely on the more representative collection of zooarchaeological records from the settlement context. Albeit somewhat earlier in time, considering the same environmental conditions and temporal proximity, we presume

Table 1 An overview of samples and types of analyses applied to Pada cemetery material covering skeletal remains and grave good pottery (cf. SI Table 1 for details). Please note that 22 additional ceramic vessels (22 ceramic powder and 7 foodcrust samples) were analysed from Pada hillforts I and II (cf. SI Table 6). SIA, stable isotope analysis; ORA, organic residue analysis

Burial no.	Sex	Age (years/months)	SIA, no. of samples	ORA, no. of samples	Microfossils (pottery)	Microfossils (dental calculus)
VI	Male	18–20	3	1	1	1
IX	Male	45–50	2	1		
XII	Child	12y ± 3 m		2	1	
XIV	Male	30–35		2	1	
XV	Child	3y ± 12 m	1	1	1	
XIX	Female	18–20?		1	1	
XVII	Female	45–50	2			
XX	Male	50–55	2			
XXIII	Male?	10y ± 30 m	2			
XXVI	Child	9y ± 24 m	3			
XXVII	Male	30–35		1	1	
XXIX	Male	30–35	4			1
XXXIII	Female	30–35	2	1	1	
XLI	Female	Undetermined	2	1	1	1
XLII	Female	25–30		2		
XLVI	Male	30–35	4	2		
L	Female	25–30		1	1	
LVIII	Female	20–25	2	1		
LXI	Male	50–55		1	1	1
LXIV	Female	25–30	4			
LXVIII	Child	12y ± 30 m		1	1	
LXXV	Female	30–35	3	1		
LXXVIII	Male	30–35	3			
LXXIX	Child	Undetermined	2			
LXXXIII	Male	40–45		2	1	
LXXXIX	Male	55–60	4			
XCII	Male	18–20		1	1	
CIII	Male	30–35	2	1	1	1
CVIII	Male	30–35	4			
CIX	Male	35–40	3	1		
CXIII	Child	11y ± 30 m	2			
CXVIII	Female	30–35	1			
CXXIV	Male	25–30		1	1	
CXXVIII	Child	9y ± 24 m		1		
CXXIX	Child	10y ± 30 m		1	1	
CLI	Undetermined; male based on grave goods	Undetermined	1			

Table 1 (Continued)

Burial no.	Sex	Age (years/months)	SIA, no. of samples	ORA, no. of samples	Microfossils (pottery)	Microfossils (dental calculus)
CLIV	Undetermined; male based on grave goods	Undetermined	3			
CLXI	Undetermined; female based on grave goods	Undetermined	2			
		Total:	**63**	**28**	**16**	**5**

that animal exploitation in the preceding (Pre-)Viking Age period largely resembles the FIA economy. The latter reasoning also applies for choosing pottery for ORA analysis. Namely, we took samples from both hillfort I and II assemblages, prioritising the sherds with foodcrust, with an end goal of making the daily dietary information more representative.

A total of 28 vessels from Pada cemetery (all foodcrust samples) and 22 vessels from the I and II hillforts (covered by 22 ceramic powder and 7 foodcrust samples) were subject to multimethod ORA analysis. The selection of pottery samples from the cemetery context was based on two aspects. Sherds with foodcrust were prioritised, for a) targeting the latest use cooking event (Miller *et al*., 2020), *i.e*. the pottery surface adhered crusts representing foods cooked before or in direct relation to funerary rituals, and b) conducting combined microfossil and lipid ORA analysis. Samples from complete vessels or rim sherds were preferred to avoid duplicate sampling of the same vessel. As the pots were concealed in a way that they could be clearly related to specific individuals, the balance and representation of pots from different age groups and sexes were also considered. For hillfort vessels, both foodcrust (if available) and ceramic powder samples were analysed to obtain an overview of major dietary sources in daily contexts.

Human remains for SIA (SI Table 2a) were selected based on having females, males, and children from different age groups, but also relying on the preservation of the skeletons and the presence of ceramic vessels in the graves. If possible, the reconstruction of dietary life histories was targeted sampling up to four skeletal elements (mostly from the mandible) from each individual: (a) M1 representing the individual's diet in the age of 1–3 years, (b) M2 in the age of 5–8 years, (c) M3 in the age of 11–14 years, and (d) mandible ca. 10 years before death (Beaumont & Montgomery, 2015; Hedges *et al*., 2007). For one child, only petrous and temporal bones were available, representing the diet in early childhood (1–3 years) and before death respectively (Jørkov *et al*., 2009). For two children, deciduous m1 was sampled, representing the diet at the age of 0.5–1.5 years (Beaumont & Montgomery, 2015).

The bone collagen SIA analysis is expected to represent the general daily diet at Pada, and in the cases where multiple bone elements were analysed, also variations in dietary intakes at different ages (lifetime dietary biographies) can be mapped.

However, the dental calculus is a more complicated sample matrix, and its formation cannot be directly correlated to any specific age and time period of an individual's life (Power *et al*., 2015; Radini & Nikita, 2023)—it can accumulate over longer and shorter time periods, sometimes be removed ((un)intentionally and mechanically) in one's lifetime, and start forming again. Without any intricate sampling strategies considering an individual's biological age and tooth formation, the dental calculus samples can only provide information on lifetime dietary practices on a very general scale. The latter is also the case in the current study, and we cannot draw any direct correlations with the bone collagen SIA results other than the broader patterns of main dietary sources. Altogether, 86 tooth/bone samples from 29 individuals were analysed in this study. Among these 29 burials, pottery ORA was conducted for 11 grave good vessels, whilst for five individuals, dental calculus microfossil analysis could be also conducted. Twenty Pada animal specimens—both domestic (cattle, sheep, goat, pig, horse) and wild (seal, beaver, and hare)—were sampled to further supplement local faunal reference baselines (Aguraiuja-Lätti *et al*., 2022) with the most abundant species evident in Pada faunal assemblages. The majority of the animal bone samples, however, did not have sufficient collagen preservation and could not be used in further analysis (for details, see SI Table 2b).

Methods

Collagen Stable Isotope Analysis

Skeletal remains were sampled, removing an outer layer of the specimen and drilling ca. 200 mg of bone powder (in three cases, fine bone fragments were used). In the case of human teeth, the junction of the tooth and crown was targeted, including the dentin under the crown. The collagen extraction was conducted at the Archemy Lab, UT, using a modified Longin method after Brown *et al*. (1988) (see details in SI Material 1). Stable carbon δ^{13}C and nitrogen δ^{15}N isotopic values were measured from duplicates using an automated carbon and nitrogen elemental analyser isotope ratio mass spectrometer (EA-IRMS) at the Department of Geology, UT, and calibrated against international standards of VPDB marine limestone standard for carbon and AIR for nitrogen (cf. SI Material 1).

Data from Aguraiuja-Lätti *et al*. (2022) and three successful animal bone samples produced within this study were used for local carbon and nitrogen baselines of terrestrial and aquatic fauna (SI Table 2b). Statistical analysis was applied to test the hypothesis of dietary differences between different sex and age groups using the Kruskal–Wallis (pairwise) statistical test for equal medians.

To quantify the food groups contributing to the diets of Pada individuals, Bayesian statistics-based FRUITS modelling (Fernandes *et al*., 2014) was applied using FRUITS 3.0 Beta desktop programme. The FRUITS model is one of the stable isotope probabilistic mixing models using the isotopic measurements of locally available food sources and their protein content as priors to better estimate the relative proportions of main dietary compositions (contributions from different food groups), building on calculations of dietary routing of different macronutrients to relevant

consumer tissues (Cheung & Szpak, 2020). Here, the three models were created, building on the information obtained from different dietary proxies and local archaeological knowledge. Aguraiuja-Lätti *et al*. (2022) datasets together with three successful animal bone samples from this study were used for isotopic baseline source values in the models. Additionally, to better estimate the proportions of C3 plant consumption and given that no plant macrofossils are known from Pada, bulk SIA measurements of major C3 cereals (wheat, barley, and rye) from the Viking Age hillfort of Iru, northern Estonia (Chen *et al*., 2023) were used as reference values for cereal consumption. The Iru cereals have moderately elevated δ^{15}N isotopic values (mean values 4.3 ± 1.8‰) for C3 crops, the tendency also detected in later materials across Estonia (Lempiäinen-Avci *et al*., 2025) indicating low to medium scale field fertilisation practices (Bogaard *et al*., 2013) in Late Iron Age and Early Medieval Estonia. All the models were restricted to mapping the contribution of three major food sources, selected according to the three dominant food groups identified among different dietary proxies. The wider aim of creating multiple models was to (a) exemplify if and to what extent the model outputs depend on the initial model input restrictions, and (b) acknowledge the possible biases of using information from a single dietary proxy for model inputs, as well as the merits of combining multiproxy dietary datasets for creating SIA mixing models. The model specifications are given in SI Table 3.

Three models tested were as follows:

- **Model A** relies on the excavation records from Pada. As plant macrofossils are lacking from archaeological records, plants are excluded from this model, and the model exclusively relies on the results of zooarchaeological analysis from Pada (Pre-)Viking Age settlement as socio-culturally and spatio-temporally closest dietary comparison applicable to Pada FIA burials. Based on the number of bone fragments assigned to different species from Pada settlement (Maldre, 2007), the proportions of animal groups are as follows: 74% terrestrial herbivores, 13% horse, 9% suids, 3% wild species (whereas the majority of them are seal bones). Only six fish bones have been identified in the assemblage. Relying on this information, the sources for Model A with their estimated contributions were assigned as: 85% terrestrial herbivores, 10% terrestrial omnivores, 5% marine fauna.
- **Model B** relies on the pottery ORA results from Pada LIA hillforts representing daily food practices (see results below, and SI Table 6). From 29 samples, potential terrestrial animal-related biomarkers (both ruminants and porcine) were detected in 14, and aquatic (fish or mammal) biomarkers in three samples. Plant biomarkers were detected in 20 samples. However, some of them are not considered reliable sole criteria for identifying plants; hence, at least two plant-specific criteria had to be fulfilled to count as plant identification, resulting in 16 samples identified as containing plants. The richness of plant microfossils detected from Pada settlement pottery serves as wider support for considerable plant consumption. The ORA results indicate that several samples present a mixture of food substances rather than a single source, and it is difficult to estimate the exact proportions of different food groups. Combining all the background knowledge and

keeping those potential biases of ORA in mind (see below), the model sources with their estimated contribution for Model B were assigned as 50% C3 plants, 40% terrestrial animals (herbivores and omnivores), and 10% marine fauna.

- **Model C** considers an overall archaeological information, incl. faunal and botanical records, also from nearby contemporaneous sites of LIA Estonia. The inclusion and proportion of C3 plants in the diet are based on the abundant finds of crop cultivation tools from LIA contexts (Lõhmus *et al*., 2011; Oras *et al*., 2018; Tvauri, 2012), but also direct evidence of plant macrofossils from contemporaneous sites (Johanson *et al*., 2024; Tvauri & Vanhanen, 2016). The estimated contributions of three major food sources for Model C were assigned as 33% C3 plants, 33% terrestrial animals (herbivores and omnivores), and 33% marine fauna.

Microfossil Analysis

A total of five dental calculus samples from five individuals were subject to microfossil analysis. The analyses were conducted at the Archaeobotany Lab, Department of Archaeology, UT. The dental calculus was kept on the tooth during the cleaning process, and a stereomicroscope with magnifications up to 110×was used during the cleaning process. As teeth had been glue-coated (identified as alcohol-soluble polyvinyl butyral (PVB glue)) in the 1980s, the first step was to remove the glue using ethanol (96.7%). After this, an established decontamination protocol was followed (Warinner *et al*., 2014; see details in SI Material 1). An acupuncture needle was dipped in weak HCl acid (0.01 M) and used to brush off and dissolve any dirt adhering to the dental calculus. Thereafter, subsamples of dissolved calculus-water and weak HCl solutions were created, and the mixture was mounted as subsamples onto microscope slides. Different teeth yielded different amounts of samples (burial no. CIII – 3; LXI – 8; XLI – 5; VI – 2; XXIX – 2). The slides were analysed using an Olympus BX51 transmitted light compound microscope with up to 500×magnification. All identifiable debris was recorded, and identification of findings was carried out using published sources and local reference collections (see SI Material 2 for details).

A total of sixteen foodcrust samples from cemetery pottery were analysed for their microfossil content (SI Table 4) at the Archaeobotany Lab, Department of Archaeology, UT. The hillfort pots had minimal foodcrust. To reduce contamination risk from the environment, the top layer was removed until no evidence of soil remained, and the deepest part of the crust was sampled for further analysis. Approximately 1 mg of foodcrust powder was mixed with ultrapure water, sonicated for 3 min, mounted on microscope slides with a drop of 1:1 glycerol-distilled water solution, sealed, and viewed immediately. An Olympus BX51 transmitted light compound microscope was used for the identification of microremains at magnifications of 200×, 400×, and 500×. The finds were documented in photographs, and the identification was based on published sources and local reference collections (see SI Material 3 for details).

Pottery Organic Residue Analysis

A total of 57 samples—28 from cemetery context (all foodcrusts) and 29 from hillforts (22 ceramic powder, 7 foodcrust samples)—were subject to multimethod ORA analysis. Foodcrust samples of ca. 20 mg were removed from the internal surface of potsherds with a clean scalpel and ground with a pestle and mortar. Ceramic samples of 0.5–1 g each were drilled using clean drill bits after removing foodcrust or 1–2 mm of outer surface of the vessel to remove possible contaminants.

Considering the limited amount of foodcrust samples and combinations of multimethod ORA and microfossil analysis, the cemetery samples were subject to acid extraction (Courel *et al*., 2020; Craig *et al*., 2013). The latter also enabled direct compound-specific stable isotope analysis of palmitic ($C_{16:0}$) and stearic ($C_{18:0}$) fatty acids with gas chromatography-combustion-isotope ratio mass spectrometry (GC-C-IRMS), which was considered crucial not only to allow better characterisation and systematic comparison of the results from different inhumations (*e.g.* sex and age-specific variations), but also to combine biomarker and isotopic results (see below). Hillfort samples were subject to solvent extraction (Colonese *et al*., 2017; Evershed *et al*., 1990, 2002) only. This approach allowed for making the hillfort pottery ORA dataset directly compatible with previously collected settlement ORA results (Chen *et al*., 2023), for which only the solvent extraction method was applied, and hence gather a more extensive and coherent dataset of pottery ORA representing daily dietary practices at Pada.

All the samples were subject to GC-FID analysis for lipid quantification, followed by GC-MS analysis under scanning and selected ion monitoring (SIM) modes. Two SIM methods targeting a) aquatic biomarkers (aqua-SIM), and b) C4 broomcorn millet-specific miliacin (miliacin-SIM) were applied to all the samples. Due to sample availability, bulk SIA using elemental analysis-isotope ratio mass spectrometry (EA-IRMS), and compound specific SIA using GC-C-IRMS was applied to cemetery foodcrust samples only. The details of sample preparation and instrument settings are presented in SI Material 1.

For plotting ORA isotopic results a dedicated graph displaying theoretical mixing lines (or curves) between major food groups was created using RStudio programme. The logic of the used R script followed the previously published work by Evans *et al*. (2023). The aim was to map the isotopic values resulting from mixing dairy products with four different food sources—marine, freshwater, ruminant adipose, and omnivore adipose fats—and build four theoretical mixing curves, *i.e.* mixing dairy with one of those food groups at 10% increments. The reference datasets and source values for these five different food groups were subtracted from Courel *et al*. (2020), using Baltic datasets only, whilst operating with mean values of isotopic measurements in the model. The estimations of concentration of different fatty acids in the five major food groups, contributing to the outline of mixing curves due to different proportions of the fatty acids in different food substances, were based on U.S. Department of Agriculture FoodData Central database (Haytowitz *et al*., 2009). The results from archaeological samples were plotted against the created mixing lines to better estimate potential mixtures of food sources in grave good vessels, whilst combining both the isotopic measurements and detected aquatic biomarkers (see below).

Results

Lifetime Diet from Stable Isotope Analysis

On visual observation, the human and animal remains did not differ in their level of preservation. Despite that, the collagen yield of the animal samples was significantly poorer than in human samples (success rate for humans 63/86 samples, 73%; for animals 3/20 samples, 15%). This might be due to the different archaeological contexts of these osseous materials: human remains come from the burials, whilst animal remains are most probably butchering and cooking waste. However, note that in Aguraiuja-Lätti *et al*. (2022), four Pada animals were successfully analysed for carbon and nitrogen stable isotopes.

Stable isotope data was retrieved for 25 inhumations (SI Table 2a), representing dietary records of 63 individuals, as different bone elements express food intake in different ages. Although different skeletal elements represent distinct age groups and are typically analysed separately (Beaumont & Montgomery, 2016; Cox & Sealy, 1997; Henderson *et al.*, 2022), our aim was to explore population-level dietary patterns rather than reconstruct individual life histories. Each measurement was therefore treated as an independent data point representing a specific age category within the Pada cohort, following the approach of Sealy *et al*. (1995) and Eriksson and Lidén (2013), where skeletal elements are considered as independent entities representative of the individual member of the population. Non-adult isotope values were integrated into the same framework as adult data, as children's diets reflect cultural norms and resource availability, making them equally informative for reconstructing community-wide practices. We fully acknowledge that our dataset is patchy, unbalanced, and not strictly normally distributed, so no statistical test fully aligns with its structure or our theoretical approach, which targets population-level patterns rather than individual life histories. However, to make maximum use of this somewhat problematic data, we decided to apply the Kruskal–Wallis test as an exploratory tool, following Sealy *et al*. (1995) and Eriksson and Lidén (2013), whilst interpreting these results with considerable caution.

All successful human and animal samples had collagen yields of over 1% and C:N ratios between 3.0 and 3.6, which indicate good quality collagen (DeNiro, 1985; for details and unsuccessful samples, see SI Table 2). In human data, the stable isotope results varied moderately across the age groups and sexes (Table 2; Fig. 2): δ^{13}C values between −22.29 and −19.11‰ (sd = ± 0.54), δ^{15}N values between 8.46 and 12.89‰ (sd = ± 1.0). These remained within a single trophic level shift (δ^{13}C range: 3.18; δ^{15}N range: 4.43) as referenced by Lee-Thorp *et al*. (2010), O'Connell (2023), Schoeninger and DeNiro (1984). The Kruskal–Wallis pairwise statistical test for equal medians comparing males and females for the δ^{13}C and δ^{15}N values did not show any statistically significant differences ($p_{\delta 13C} = 0.3025$ and $p_{\delta 15N} = 0.8162$), *i.e.* there was no major dietary differences between the sexes. A few outliers emerged: male burial no. CLI sample M3 (representing diet at 11–14 years) had higher δ^{15}N value (12.85‰), and male burial no. CVIII sample M3 (11–14 years) had lower δ^{13}C value (−22.29‰) than the rest of the population. The only outlier among females

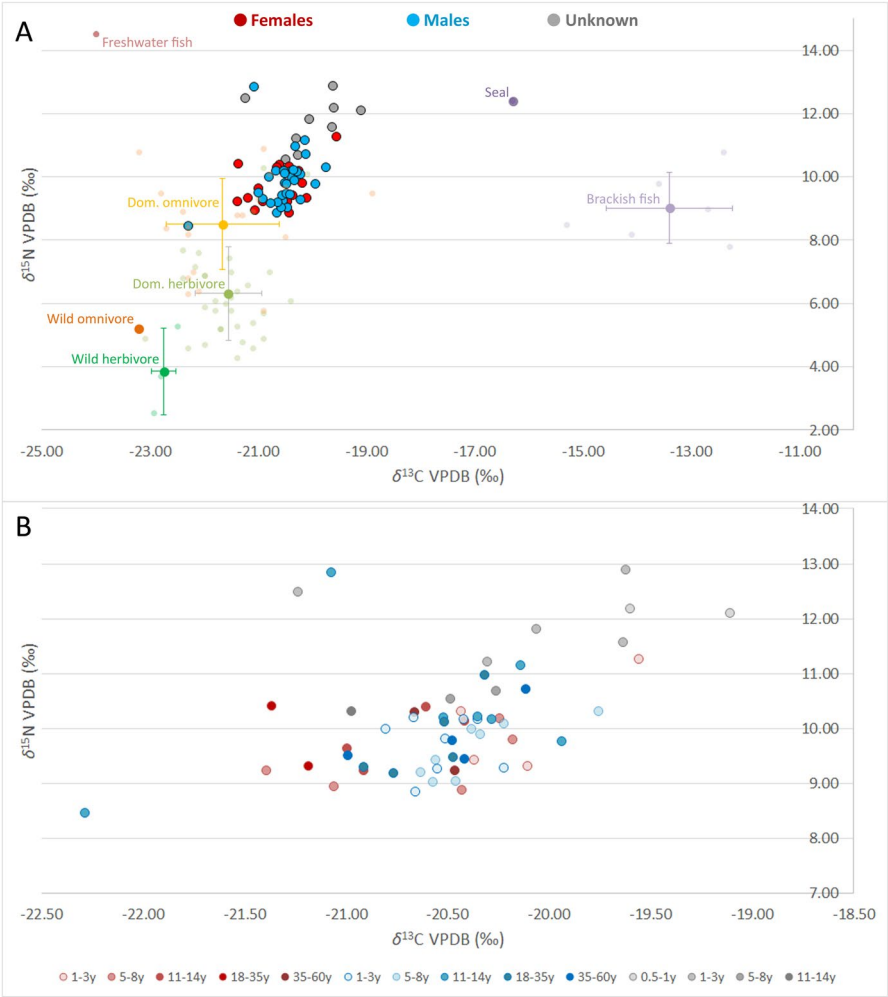

Fig. 2 SIA results for the extended population in Pada ($n = 59$). A total population of analysed individuals with **A** osteological sex estimation; **B** sex and age estimation, where colour saturation indicates age groups, *i.e.* darker—older; lighter—younger. Animal isotopic data derives mostly from Aguraiuja-Lätti *et al*. (2022) and three additional animal bone samples reported in SI Table 2b

was a probable 1–3-year-old of burial no. CLXI with higher δ^{15}N value (11.27‰), most likely reflecting breastfeeding (Fogel *et al*., 1989; Fuller *et al*., 2006; Morrone *et al*., 2023).

No major dietary differences were detected across age groups either. The Kruskal–Wallis test for equal medians among the six assigned age groups (SI Table 5) detected some differences in δ^{15}N values between age groups 0 and 2 ($p = 0.0254$), 0 and 4 ($p = 0.0404$), and 0 and 5 ($p = 0.0455$). These differences could be attributed to breastfeeding in group 0 with ca. 1‰ higher δ^{15}N values in general (Fogel *et al*., 1989; Fuller *et al*., 2006). For δ^{13}C the statistically

Table 2 $\delta^{13}C$ and $\delta^{15}N$ results for males ($n=32$, M), females ($n=17$, F), and non-adults of unknown sex ($n=10$, UK) in different age groups. The dataset covers the so-called extended population representing 63 individuals sampled from 25 inhumations. Individuals with unknown sex and age ($n=4$) are not shown

Sex	Age group (years)		No. of ind	$\delta^{13}C$								$\delta^{15}N$							
				Min	**Max**	**Range (‰)**	**Mean (‰)**	**SD**	**Median**	**Q1**	**Q3**	**Min**	**Max**	**Range (‰)**	**Mean (‰)**	**SD**	**Median**	**Q1**	**Q3**
M	1	1–3	8	−20.81	−20.22	0.58	−20.52	0.19	−20.53	−20.66	−20.40	8.86	10.21	1.35	9.73	0.52	9.91	9.29	10.18
	2	5–8	8	−20.63	−19.76	0.88	−20.37	0.28	−20.42	−20.56	−20.31	9.03	10.32	1.29	9.63	0.51	9.66	9.17	10.02
	3	11–14	7	−22.29	−19.94	2.35	−20.66	0.80	−20.35	−20.80	−20.21	8.46	12.85	4.39	10.41	1.34	10.20	9.98	10.69
	4	18–35	5	−20.91	−20.32	0.59	−20.60	0.24	−20.52	−20.77	−20.48	9.18	10.98	1.80	9.82	0.75	9.48	9.31	10.13
	5	35–60	4	−20.99	−20.12	0.88	−20.50	0.36	−20.45	−20.61	−20.34	9.45	10.73	1.28	9.87	0.59	9.65	9.50	10.02
F	1	1–3	4	−20.44	−19.56	0.88	−20.12	0.40	−20.24	−20.39	−19.97	9.33	11.27	1.94	10.09	0.91	9.88	9.40	10.56
	2	5–8	5	−21.39	−20.18	1.21	−20.66	0.54	−20.43	−21.06	−20.25	8.88	10.19	1.31	9.41	0.57	9.23	8.94	9.81
	3	11–14	4	−21.00	−20.42	0.58	−20.73	0.27	−20.76	−20.94	−20.21	9.23	10.39	1.16	9.85	0.52	9.89	9.54	10.20
	4	18–35	2	−21.37	−21.18	0.18	−21.28	0.13	−21.28	−21.32	−20.48	9.33	10.41	1.08	9.87	0.76	9.87	9.60	10.14
	5	35–60	2	−20.66	−20.46	0.20	−20.56	0.14	−20.56	−20.61	−20.51	9.24	10.31	1.07	9.78	0.75	9.78	9.51	10.04
UK	0	0.5–1.5	2	−19.60	−19.11	0.49	−19.36	0.35	−19.36	−19.48	−19.23	12.10	12.18	0.08	12.14	0.06	12.14	12.12	12.16
	1	1–3	5	−21.24	−19.63	1.61	−20.17	0.66	−20.06	−20.31	−19.64	11.21	12.89	1.68	12.00	0.68	11.82	11.57	12.48
	2	5–8	2	−20.49	−20.26	0.22	−20.38	0.16	−20.38	−20.43	−20.32	10.55	10.69	0.14	10.62	0.10	10.62	10.58	10.66
	3	11–14	1	−20.97	−20.97	0.00	−20.97	n/a	−20.97	−20.97	−20.97	10.32	10.32	n/a	n/a	n/a	10.32	10.32	10.32

significant difference was found between group 0 and all the other age groups, and between groups 1 and 4 ($p = 0.0284$). Explanation of the differences of median $\delta^{13}C$ values between group 0 ($\delta^{13}C_{mean} = -19.36‰$) and other age groups, with group 0 being more enriched, is challenging but probably has something to do with the female/mother's diet. The difference for groups 1 ($\delta^{13}C_{mean} = -20.33‰$) and 4 ($\delta^{13}C_{mean} = -20.79‰$) is less than a per mil. However, we want to emphasise that all these comparisons must be interpreted cautiously due to small sample sizes (limited individuals) forming those age groups and given that our dataset does not fully adhere to necessary criteria for more rigorous statistical comparisons.

The detailed results of the FRUITS models are presented in SI Table 3. For the sake of clarity, the averaged results across two adult biological sex categories and three non-adult age groups are presented here (Fig. 3). All the models highlight the main dietary tendencies rather clearly. First, the proportion of aquatic/marine diets is minimal (<3%). Depending on whether plants are included in the model or not, major food substances are formed by either C3 plants (77–84%, Models B and C) or terrestrial herbivores (85%, Model A). Model A (based on zooarchaeological records, yet lacking plants), however, clearly omits one crucial dietary source—plants—entirely. When C3 plants are included in the model, the proportion of terrestrial animals falls within the range of 15–21%. The omnivores seem to form a minor part of subsistence (12%, Model A). It is noteworthy that in our reference baseline dataset the difference between these terrestrial omnivores and herbivores groups is minimal (1–2‰ for both C and N) and hence the clear distinction of the two might be arbitrary anyway. There are no sex-based dietary variations in the FRUITS models either.

For non-adult groups, the proportions of main food sources are the same by and large, with the only exception of up to 4% possible marine consumption, and a somewhat higher percentage of omnivorous food (11–16% in Model A) if separated out in the model. Otherwise, the model shows clear reliance on C3 plant-based foods across different age categories among children. Comparing the three non-adult age groups with one another, there is a significant difference in Model A (marine; $p = 0.04061$) and Model B (all food groups; $p = 0.04411$) between the age group of

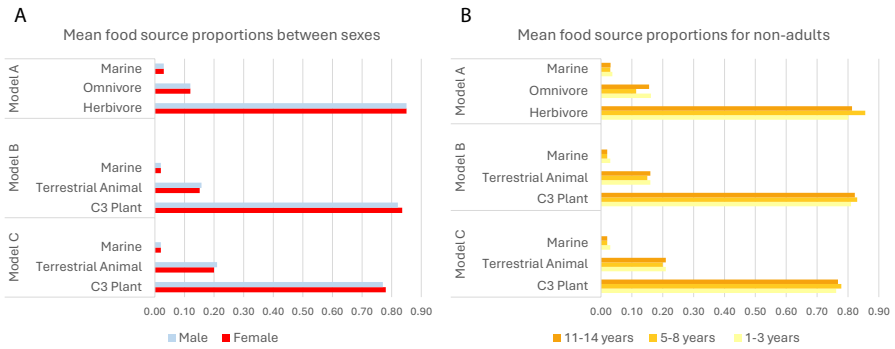

Fig. 3 Proportions of different food sources according to the three FRUITS models of Pada SIA results: **A** adult comparison by sex and **B** children comparison by age group

1–3-year-olds and that of 5–8-year-olds, as indicated by the Kruskal–Wallis test of the equal medians (see SI Table 3). This could indicate the breastfeeding effect of the younger group. We do not observe any statistical differences between the male and female groups, nor whilst comparing these with the non-adults at the age of 5–8 years, meaning that children and adults had access to the same foods.

Considering that Model A is biased in omitting C3 plants from the dietary intake entirely, the overall interpretation of lifetime diet ought to rely on Models B and C. The latter show that regardless of sex and age groups, the Pada community had C3 plants as their main staple (> 75%), followed by terrestrial animals covering both omnivores and ruminants (> 15%), whilst marine resources formed only a couple of percentages of the overall food intake. However, it must be noted that the proportion and estimation of marine resources purely rely on the fact that we have evidence for marine mammals in our zooarchaeological assemblage, whilst any fish (either marine or freshwater) records are currently missing. The latter may hinder the visibility of other aquatic resources in our models, but without extensive (alternative) records proving the consumption of freshwater resources, we cannot factor in the contribution of this particular dietary source in our models.

Dental Calculus Microfossil Records

Dental calculus analysis revealed multiple microfossil categories (Table 3) from different kingdoms of life (Animalia, Fungi, and Plantae). Starch was found from samples of all individuals. Most numerous were unidentified starch granules ($n = 17$); notable findings include Triticeae ($n = 5$) and Fabaceae ($n = 4$) starches; one possible

Table 3 Microfossils identified from Pada dental calculus samples. A –all microdebris categorised by type. B – identified starch findings. Roman numbers indicate burials/individuals

A	**CIII**	**LXI**	**XLI**	**VI**	**XXIX**	**Total**	**% of total**
Animalia	32	45	9	2	8	96	9.9%
Bacteria	0	1	1	0	0	2	0.2%
Fungi	4	86	6	2	2	100	10.3%
Non-living nature	17	42	29	8	4	100	10.3%
Plantae	24	47	27	10	7	115	11.8%
Problematic	88	351	80	18	24	561	57.6%
						974	
B	**CIII**	**LXI**	**XLI**	**VI**	**XXIX**	**Total**	**% of total**
Starch	2	1	2			5	0.5%
Cf. starch	2	5	7	1	2	17	1.7%
Legume starch		1				1	0.1%
Cf. legume starch				3		3	0.3%
Cf. *Avena* starch	1					1	0.1%
Triticeae starch	1	1	2		1	5	0.5%
Cf. tuber starch	1	1				2	0.2%
Total	7	9	11	4	3	34	

Avena starch granule was found as well. Tuber starches ($n=2$) were not identified with certainty to any taxa, but other possible sources for starches were identified (Fig. 4). Kingdom Animalia includes epithelial cells, which are commonly found in the lining of human mouths (Tinanoff & Gross, 1976, see examples in Blondiaux & Charlier, 2008)—these are often numerous in dental calculus analyses and do not necessarily hint at dietary input. Bone/meat debris was also identified as clear indicators of dietary practices. Fungal debris was numerous ($n=100$). Nine of these were identified as Glomeromycota, of which fungal fruiting bodies are commonly found within sediment. This could indicate uncleaned food or contamination from the burial setting. Identification groups "non-living nature" and "problematic debris" consist of debris that could not be associated with any other kingdom, such as minerals for the former or burnt debris for the latter. All in all, dental calculus exhibits the consumption of starchy plants, covering both cereals and legumes, but also tubers and meat-based foods, whilst the findings of both plant- and animal-related groups have rather equal proportions. A more detailed description of all mentioned categories and possible pathways of the microdebris inclusion is reported in SI Material 2.

Microfossil Records from Pottery Foodcrusts

The observed microfossils in the cemetery pots are classified into several categories (Fig. 5, Table 4; for details see SI Material 3). The majority of quantitatively identified microfossils were phytoliths, but there were also starch grains, pollen

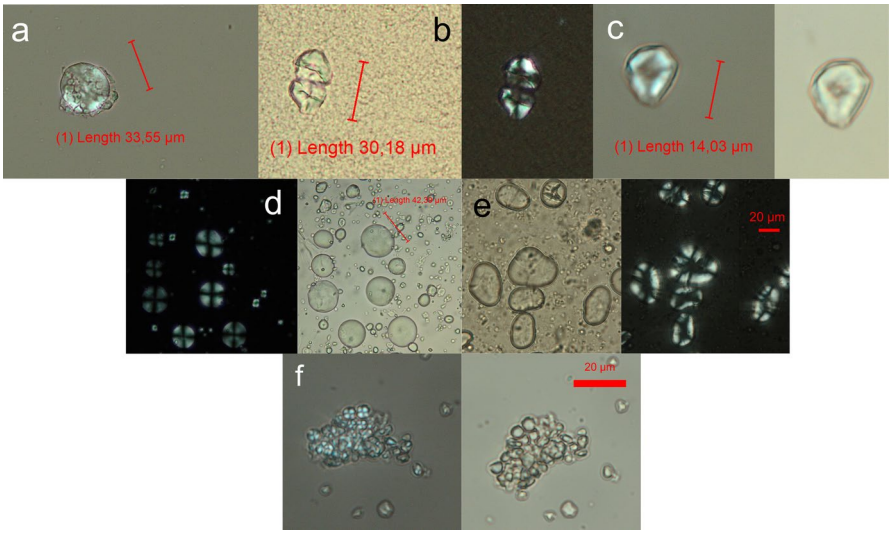

Fig. 4 Starch from Pada dental calculus: **a** Triticeae-type starch found from burial no. XLI. Notice the A- and B-type starches characteristic of the Triticeae tribe. **b** Fabaceae starch from burial no. VI. Notice the distinctive 'arms' of the extinction cross that sprout unevenly from the centre. **c** cf. *Avena* starch from burial no. CIII. Comparative material from reference collections: **d** *Secale cereale*, **e** *Pisum sativum*, and **f** *Avena sativa*. Notice the angular shape and indefinite extinction cross on some granules

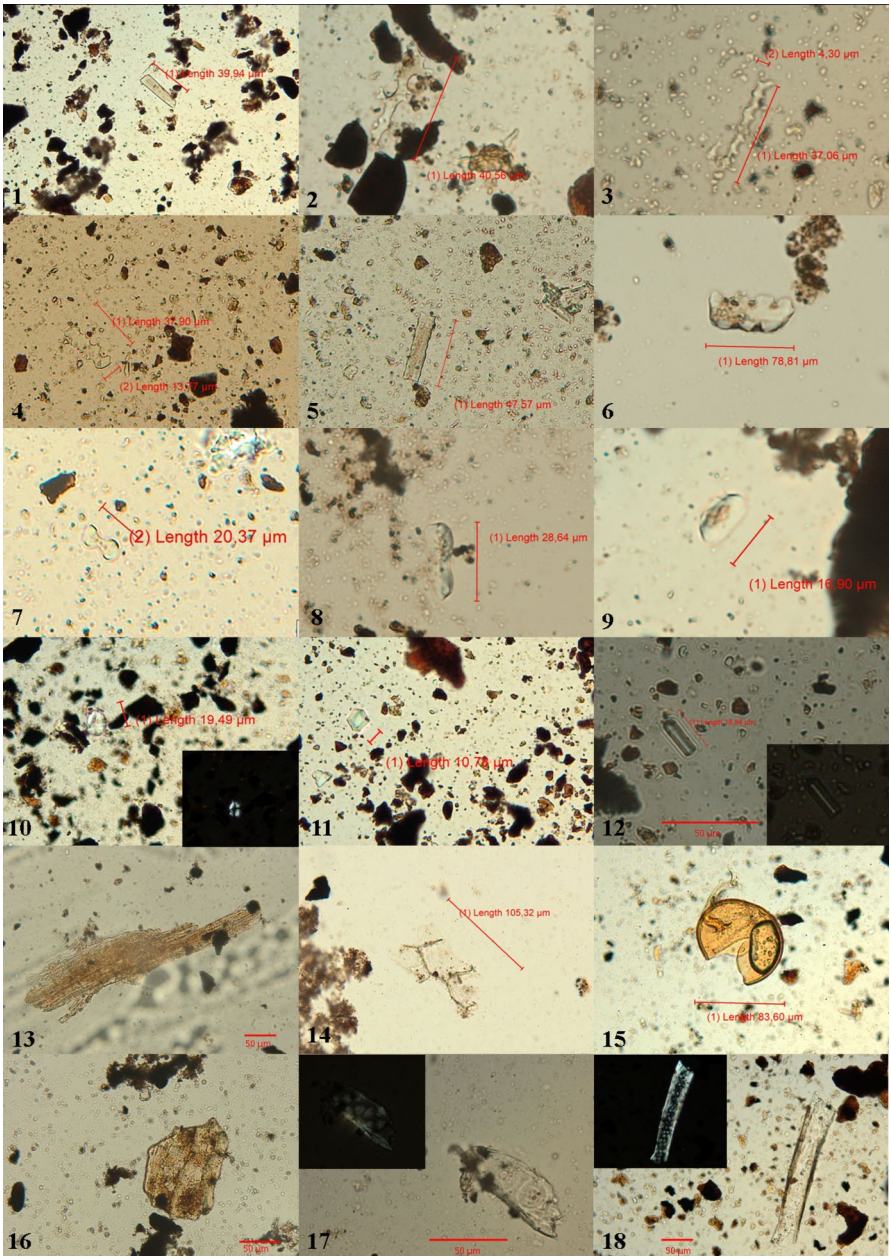

Fig. 5 Microremains from Pada foodcrust samples. 1–5 elongated types (1—psilate, 2, 3—dendritic, 4—sinuate, 4—verrucate); 6—trapezoid sinuate, 7, 8—bilobate, 9—rondel, 10—starch, 11—calcium oxalate blocky (food crystal), 12—calcium oxalate prismatic, 13—plant tissue, 14—plant cell, 15—Glomeromycota t. fungal spore, 16—cf. fish, 17, 18—conifer wood remains

grains, calcium oxalate crystals, wood tissue, sclereids, and other plant tissue (both

processed and/or burnt plant matter as well as plant cells). Other types of micro-remains included diatoms, fungal remains (esp. spores and hyphae of Glomeromycota type fungi), possible bone and flesh remains, oil bubbles, and insect particles. Numerous crystals, possibly calcium-based crystals formed during food processing events, like mashing and fermenting (Dietrich *et al*., 2012; Heiss *et al*., 2020) were also discerned, but their exact nature is uncertain, as they can be fermentation products of animal-based foods (*e.g.* milk) or originate from plants as well.

Twenty-two different morphotypes were distinguished among phytoliths (see SI Table 4). Of these, elongated dendritic, rondels and trapeziform are most likely associated with consumed cereals (Barboni *et al*., 2007; Danu *et al*., 2019; García-Granero *et al*., 2021), whilst bilobates could refer to Panicoidea (García-Granero *et al*., 2021). The scarcity of silica skeletons as well as very fragmented phytoliths refer to the heavy processing (*i.e.* grinding) of grains. The presence of cereals is also indicated by three possible B-type of Triticeae starches. The pot with the most cereal-associated phytoliths was from the burial LXVIII.

Wood remains were found from all analysed samples, several of which could be identified as conifers by distinctive wood pits, cubic phytoliths (Gao *et al*., 2018), and calcium oxalate crystals. The presence of conifer wood in the pots could be explained by the wide application of conifers as heating fuel for cooking. Wood residue could also refer to containers' lids or ladles or edible inner bark (Fjellström, 2020; Kalle *et al*., 2013).

The hyphae and spores of Glomeromycota type fungi are extremely numerous in some samples (60 in sample from burial LXXXIII; 53 in XCII, 45 in LXVIII) and could refer to cooking insufficiently washed vegetables. Small debris of processed organic matter, sometimes burnt, was found as well. In 117 cases, the cell structure of plant epidermis could be followed. In 226 cases, potential flesh or bone remains were identified, although this is a very tentative and cautious estimation, because the research on the microdebris of flesh/bone is limited and no actual muscle fibres analogous to reference material could be identified. The presence of flesh in the foodcrusts, however, can also be indicated by the oil bubbles, although their number was quite small.

Pottery Organic Residue Analysis from Hillfort and Grave Good Vessels

Lipid concentrations from hillfort samples ranged from 7 to 6048 μg g^{-1} for ceramic powder and 56 to 10,814 μg g^{-1} for foodcrusts. The details of identified compounds are presented in SI Table 6. As more diagnostic biomarkers, cholesterol and its derivatives are presented in 17 samples, whilst plant-related phytosterols (β-sitosterol, campesterol, and stigmastanol) were found in 16 samples. A total of 20 samples contain the combinations of different criteria of plant-based food, incl. saturated and unsaturated $C_{18:1}$ and $C_{18:2}$ fatty acids together with hydroxy fatty acids and dicarboxylic acids suggesting plant oils, and long-chain n-alkanes, n-alkanols, and even-numbered long-chain fatty acids suggesting plant waxes (Cramp *et al*., 2011; Evershed *et al*., 1991; Roffet-Salque *et al*., 2017), further supported by plant sterols and by C_{18} ω-(o-alkylphenyl) alkanoic acids (APAA) E/H isomer ratio > 4. Clear aquatic biomarkers, APAAs with carbon

Table 4 Microfossils identified in Pada cemetery foodcrust samples. A – total number of microremains. B – microremains most likely indicative of consumed cereals. Roman numbers indicate burials where the pots were found in

Sample ID	VI	XII	XIV	XV	XIX	XXVII	XXXIII	XLI	L	LXI	LXVIII	LXXXIII	XCII	CIII	CXXIV	CXXIX	SUM
A																	
Plantae (Total)	50	25	70	117	47	68	72	62	82	30	119	93	82	14	106	63	1100
Phytoliths	36	18	54	54	31	54	43	38	63	13	106	27	27	8	85	32	689
Starch				1							1	1					3
Oxalates	4	3	10	58	3	8	13	4	10	3	3	40	50			8	217
Pollen			1						1		1	2			3	1	9
Wood	3	3	1	4	3	4	1	3	4	4	6	8	1	1	12	7	65
Processed plant remains	7	1	4		10	2	15	17	4	10	2	15	4	5	6	15	117
Animalia	20	16	17	11	38	16	13	35	6	13	3	10	7	13	6	2	226
Fungi	16	21	25	30	42	14	8	61	22	14	76	96	101	19	23	32	600
Chromalveolata							1										1
Other (oil)						7	4	1	17								29
B																	
Phytolith morphotype																	
Elongated dendritic: grasses, likely cereals (inflorescence)	1		1	14	1	2	1	1	1		11	1	1		2	2	39
Rondel: grasses, esp. Pooidea (stems, leaves)	1	3	7	9	1	6	22	8	7	1	23	5			4	8	105
Trapezoid: grasses, esp. Pooidea (leaves, inflorescences)	1	1	2	5		3	1	1	3	4	8	2			12	6	49
Bilobate: grasses, esp. Panicoidea (leaves)			1					1	1		1				2		6

atoms ranging from C_{16}/C_{18} to C_{20}/C_{22}, together with one or more of the isoprenoid fatty acids (Hansel *et al.*, 2004), and in the majority of cases further supported by phytanic acid diastereomers (3S,7R,11R,15-phytanic acid (SRR) and 3R,7R,11R,15-phytanic acid (RRR) with SRR% > 75.5 (Lucquin *et al.*, 2016); or APAA $C_{20}/C_{18} > 0.06$ (Bondetti *et al.*, 2021)) were found in only three vessels. The ORA results from Pada hillforts resemble the analytical results from Pada settlement (Chen *et al.*, 2023), with the presence of plant-based foods in over half of the samples, and a limited number of samples with aquatic biomarkers.

Lipid concentrations from cemetery pottery foodcrusts ranged from 38 to 6724 μg g^{-1}. The details of identified compounds are presented in SI Table 6. Cholesterol and its derivatives are evident in all the samples. Indications of plant substances, as explained above, are evident in 14 samples, although in the majority of cases only one plant-specific biomarker has been fulfilled, and hence the substantial inclusion of plants remains questionable. Miliacin (Heron *et al.*, 2016) was detected in four cemetery vessels. Differently from hillfort material, a total of 19 samples include aquatic biomarkers, although the phytanic acid diastereomers SRR% ranges are relatively low (75.5–85.8; SRR% range > 75.5 indicative of an aquatic source).

The GC-C-IRMS results from grave good pottery showed the presence of dairy, ruminant, porcine, and marine substances (Fig. 6), but no freshwater food (which could overlap with potential C3 plant isotopic values, if those were present). The lack, or rather limited inclusion, of plants in the cemetery context is also evident from bulk SIA analysis of foodcrusts, where the C/N ratio remains in the range of 6–13.5 (only one sample has C/N ratio 27.2) and all the samples display $\delta^{15}N > 8.9$ ‰ (Fig. 6: A), indicative of animal-based substances (Heron & Craig, 2015; Yoshida *et al.*, 2013). Based on $\Delta^{13}C$ value of lipid CSIA measurements (> 3.1), dairy is present in six vessels. However, the majority of the samples plot outside the main reference ellipses (Fig. 6: B), instigating the presumption that we are dealing with mixtures of food sources. The graph displaying theoretical mixing curves of five different food groups (Fig. 6: C) exemplifies the presence of some clearly single compound vessel contents, but also combinations of various food substances. The latter is also supported by variations of different biomarkers in the cemetery samples, particularly a considerable number of aquatic biomarkers and the fact that several more of those RRR/SRR values score at the threshold of aquatic vs. ruminant carcass fats (see SI Table 6 for full overview of ORA results). Hence, some of the samples plotting into ruminant carcass fat range might be a result of mixing dairy with non-ruminant (porcine, aquatic) substances.

The Kruskal–Wallis test for equal medians comparing $\Delta^{13}C$ values across pots found in male ($n = 13$), female ($n = 5$), and child burials ($n = 6$) showed that there is no significant difference between the contents of the pots in these three groups ($p = 0.3347$).

Discussion

The importance of incorporating multiple dietary proxies and analytical methods for more cohesive dietary reconstructions in archaeology has been highlighted by previous scholars working with *e.g.* pottery ORA (Kooiman *et al.*, 2022; Kubiak-Martens

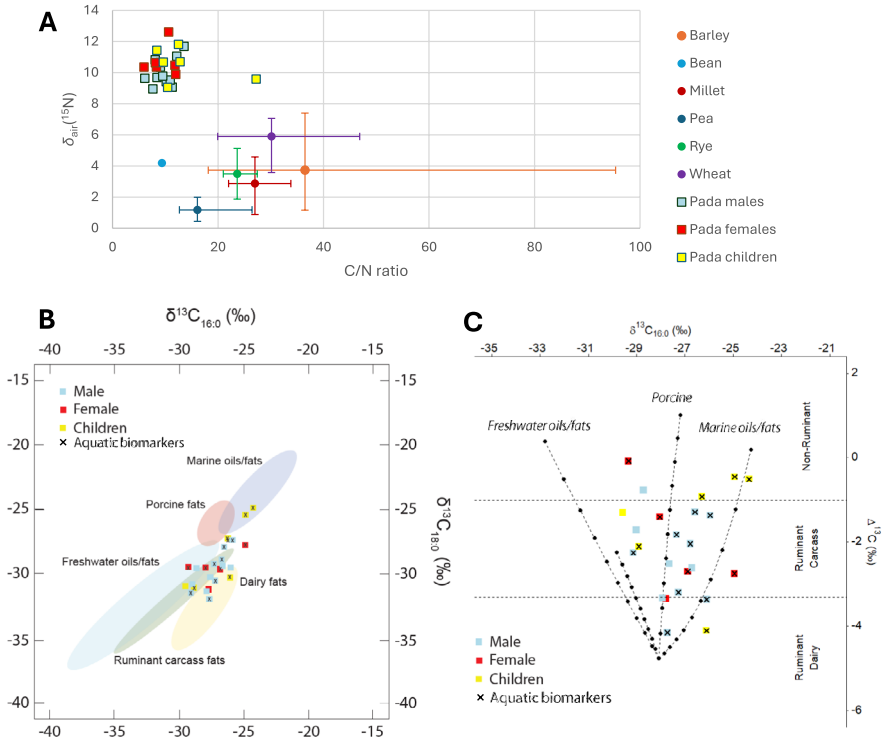

Fig. 6 ORA results from Pada grave good pottery foodcrusts. **A** bulk stable isotope results with cereal isotopic data from Chen *et al*. (2023); **B** compound specific $\delta^{13}C$ values of $C_{16:0}$ and $C_{18:0}$ fatty acids. The reference ranges are calculated at 68% confidence using datasets mostly for the Baltic region (Courel *et al*., 2020; Dudd, 1999; Pääkkönen *et al*., 2020) **C** theoretical mixing curves of $\delta^{13}C_{16:0}$ and $\Delta^{13}C$ values resulting from mixing dairy with freshwater oils/fats, porcine fats, ruminant carcass fats and marine oils/fats (source code from Evans *et al*., 2023, modern reference material from Courel et al., 2020, limited to Baltic samples), where black circles represent percentage contributions in 10% increments between the mean values. Symbol × marks the samples that included aquatic biomarkers

et al., 2024; Lucquin *et al*., 2023), human remains (Cristiani *et al*., 2018; Millard *et al*., 2020; Pedergnana *et al*., 2025), but also combinations thereof (Cramp *et al*., 2014; Oras *et al*., 2018, 2023; Robson *et al*., 2019). These works have not only proven the merit of combined analytical procedures, but also the capacities of and discrepancies between different methods.

Comparing the different analytical methods applied to Pada material in this study, the analytical and taphonomic biases of different proxies, relevant to any spatio-temporal context, become clearly visible (Table 5). The detection and, more importantly, the estimation of the proportions of plant-based foods in the daily diet tend to be underrated with bulk SIA. This is partially because human bone bulk collagen SIA reflects protein intake, and most plants have a rather low protein content (Ambrose & Norr, 1993; Lee-Thorp, 2008). The support for including plants (and their rough proportions) into modelling approaches such as FRUITS ought to derive from other proxies and methods. If the excavations did not involve sediment

Table 5 Overview of main informative capacity of different analytical methods and proxies used in ancient dietary analysis and applied to Pada material

	Skeletal bulk SIA	Bayesian modelling of SIA (*e.g.* FRUITS)	Microfossils	Pottery ORA	Ecofacts without floatation	Ecofacts with floatation
Plants						
C3 cereals	**	***	***	**	*	***
C4 plants	***	***	***	***	*	***
Legumes	**	**	**	*	*	**
Vegetables & tubers	-	-	**	**	-	**
Berries	-	-	**	**	-	**
Fungi	-	-	***	-	-	-
Aquatic resources						
Marine animals	***	***	**	***	*	***
Freshwater animals	***	***	**	***	*	***
Terrestrial animals & their products						
Ruminants	***	***	*	***	***	***
Omnivores	***	***	*	***	***	***
Dairy	-	-	-	***	-	-

***Highly informative; **some information; *limited information; - no information. *SIA* stable isotope analysis; *ORA* organic residue analysis

sampling and floatation, the macrofossil record of plants is oftentimes missing, making the estimation of plant-based foods more complicated, whilst their entire exclusion would be simply erroneous (compare *e.g.* different results in our Models A–C, Fig. 3). This information can be retrieved from microfossil analysis of either dental calculus or from a multimethod study of pottery ORA, especially when combining lipid and microfossil analysis. In principle, microfossil analysis can also be useful for widening the range of different plant groups incorporated into past dietary practices, *e.g.* legumes, vegetables, berries, and fungi, only a narrow selection of which could be detected with lipid-based pottery ORA. Although not necessarily highly diagnostic when it comes to species identification, the ORA, on the other hand, can reveal a wider array of major food groups and also detect lower inclusion of aquatic resources that do not necessarily emerge from bulk SIA measurements. In our suite of analytical approaches, ORA is also the only method capable of identifying dairy consumption. Finally, the detection of terrestrial and aquatic animals is possible with all the methods, whilst the separation of ruminant and omnivore tissues is feasible with both ORA and SIA. The major flaw of all the methods used in analysing Pada material is that the more detailed identification of exact species, including the distinction of wild and domesticated animals, remains largely unachievable. This could be addressed with more recent developments of high-resolution techniques like shotgun proteomics in the future, as these new analytical developments might be able to

distinguish various food groups, even down to the level of species (Hendy, 2021). However, due to several analytical and preservation biases, the detection of certain food groups like plants and fish is problematic even here (Evans, 2025; Hendy *et al*., 2018), especially when analysing foodcrusts that are modified by high temperatures and direct contact with the surrounding depositional environment (Dekker *et al*., 2025; Evans *et al*., 2024).

Zooming into the more detailed results from Pada, it is evident that none of the biomolecular datasets overlap entirely (Fig. 7) and information obtained from single method (skeletal SIA, pottery ORA, or microfossil analysis) or single proxy (pots, skeletal remains, ecofacts) will always be partial. Furthermore, as we have proven with the Pada example, these biases can be further complicated by socio-cultural constituents like variations of dietary practices at different sites and in different consumption contexts.

In Pada, we see heavy reliance on plant-based foods in daily dietary practices based on the FRUITS SIA models and dental calculus analysis (Fig. 7: A). However, the high proportion of plants in the overall lifetime diet would have been missed if we had relied on bulk SIA results only, *i.e.* without applying any modelling approach. The discrepancies between the modelled SIA and dental calculus datasets,

Fig. 7 Comparative overview of the main analytical findings of lifetime dietary practices from human skeletal remains (**A**) and contents of grave good pottery (**B**) from Pada cemetery materials. The SIA plot is based on the results of FRUITS model B; microfossil graphs represent the proportional division of microfossils limited to identification groups of plants or animals; the ORA plot represents the proportional distributions of summed supportive molecular criteria diagnostic to the five major food groups (based on SI table 6). SIA, stable isotope analysis; DC, dental calculus; ORA, organic residue analysis; FC, foodcrust

with the first showing higher reliance on plants than the latter, might be due to analytical biases of dental calculus. Indeed, it has been reported that plant microfossils tend to be underrepresented in dental calculus analysis (Radini & Nikita, 2023). The high proportion of the so-called "problematic debris" referring to unidentifiable burnt organic matter constitutes over half of the microfossils in our calculus dataset. These remains can be evidence of burnt food (and plants) or fuels used in heat processing of foods, but we simply lack the analytical capacity to identify them any further. Although our dental calculus dataset is limited to only five individuals, it is reassuring that all the so-called lifetime dietary proxies show high inclusion of plant-based foods. The latter is also supported by pottery ORA from habitation contexts, where plant biomarkers were detected in over half of the samples.

Our SIA modelling results show that the Pada community had C3 cereal-based diets as their main staple, with terrestrial animals and aquatic (probably marine) sources forming 10–20% and < 5% respectively. However, we must acknowledge that the biggest impact on these proportional estimations of different food groups derives not only from the modelling interface (Cheung & Szpak, 2020), but also from the used reference datasets/source values and priors (Schulting *et al*., 2022). For example, in our case, the terrestrial omnivore values had only 1–2‰ δ^{15}N difference from herbivores, which seems to hint that their diet was rather like humans, as expected for domestic pigs (Hu *et al*., 2009). The baseline δ^{15}N isotopic values of C3 plants in our models were set somewhat higher, as we have hints that LIA communities were incorporating at least some manuring into their crop cultivation (Bogaard *et al*., 2013). These baseline source values had a major impact on our modelling results, including difficulties in clear differentiation between some food groups. Hence, local and ideally also spatio-temporally compatible isotopic reference datasets for both animals and plants are crucial for more accurate estimations of dietary variations among past populations. To overcome the challenges of separating certain food groups that are partially overlapping in their reference values, a more sophisticated approach of compound specific isotopic measurements of single amino acids from bone collagen could be employed in the future (Vane *et al*., 2025). The latter in combination with Bayesian-based modelling could potentially achieve more precise estimates of the proportions of different foodstuffs, including a better separation of plant-based foods (Moubtahij *et al*., 2024; Soncin *et al*., 2021).

Tuning into the results from grave good pottery some intriguing deviations emerge (Fig. 7: B). On one hand, the comparison of pottery ORA and microfossil records from the same foodcrusts highlight the underrepresentation of plant substances in ORA results. Unless dealing with high concentrations of plants with significant amounts of oils or waxes, plants are more difficult to detect with lipid-based ORA (Dudd *et al*., 1998; Evershed, 2008a, 2008b; Hammann & Cramp, 2018; Roffet-Salque *et al*., 2017). Employing targeted analysis of certain plant biomarkers, which often tend to be present at lower concentrations, *e.g.* miliacin (Heron *et al*., 2016; Lundy *et al*., 2024), with GC-SIM analysis might be one solution. Indeed, using this method, we were able to show the inclusion of millet, albeit in small contents, in four Pada grave good vessels. However, this is clearly a rather narrow approach targeting only one compound (and type of plant). There are some more promising recent developments for developing dedicated ORA methods for detecting wider range of

plants (Bondetti *et al*., 2021; Colonese *et al*., 2017; Doliente *et al*., 2024; Heron *et al*., 2016), but until they become routine analytical practice, the incorporation of microfossil analysis is crucial for providing a better estimation of plant inclusion in pottery food residues. The microfossil results from Pada show considerable plant inclusion with clear hints of at least some highly processed cereal consumption. Starches were scarce, which is logical given the heating during crust formation resulting in starch degradation (Crowther, 2012). Comparing the efficiency of these two methods, our ORA results seem to have "missed" plants in five vessels out of 16, which is most likely due to the inclusion of lipid rich animal products dominating the composition of these residues. Indeed, the foodcrust bulk SIA measurements show dominance of higher trophic level animal-based foods in Pada grave good pots as well (Fig. 6: A). However, due to diagnostic limitations of microfossil analysis when it comes to animal tissues, the ORA clearly stands out for its capacity of identifying a wider range of different food components. All in all, our comparative analysis provides clear arguments for combined ORA and microfossil analysis to obtain a more comprehensive overview of different food components consumed or cooked in ceramic vessels.

Although the underrepresentation of aquatic foods (including its minor proportions in our SIA FRUITS models) could derive from excavation techniques and sampling strategies, overlooking smaller bone specimens, the aquatic biomarkers in hillfort (and settlement) pottery were still limited to ca. 10% of the vessels. This supports the notion that aquatic foods did not play a significant part in Pada LIA foodways. Contradictory then is the observation that most of the grave good pottery had aquatic biomarkers in them. The explanation might be twofold. First, grave good pottery reflects socio-culturally determined dietary practices, with a preference for terrestrial and aquatic animal based foods in a ritual context. Similar differentiation of ritual and daily foods has been established in the eastern Baltic (Oras *et al*., 2018) and elsewhere (Craig *et al*., 2015; Dunne *et al*., 2018). Alternatively, we might be evidencing some analytical biases here. Acid extraction (used for grave good pottery) could be more prone to yielding aquatic biomarkers, especially considering that APAAs are considered bound fatty acids, as opposed to solvent extraction (applied for hillfort pottery) (Correa-Ascencio & Evershed, 2014). Further analytical work is needed to better estimate this potential methodological bias (Whelton *et al*., 2021) in the future.

The pottery ORA results also showed that Pada LIA people had the tendency to mix different food sources in their ceramic containers in both daily and ritual context (see SI Table 6). Furthermore, the plotting of lipid CSIA measurements on theoretical mixing lines further complemented by aquatic biomarker demarcation (Fig. 6: C) serves as a methodologically warning example of how food components could be easily misassigned without combined multianalyte/-method approach. *E.g.* most of our grave good pottery samples plot into ruminant dairy or carcass region, or outside the clear reference ellipses (Fig. 6: B). Our graph with mixing lines using local isotopic baseline values (Fig. 6: C) highlights the possibility that these samples could be in fact mixtures of dairy with *e.g.* marine or freshwater sources (where aquatic biomarkers are present), or with porcine fats. These results agree with some previous work (Hendy *et al*., 2018) implying the importance of more complex mixing and modelling approaches, especially where mixed food sources of varying lipid contents and isotopic values are expected.

Turning from methodological challenges to archaeological relevance, we can conclude that by using our multiproxy and -analyte approach we have managed to create a more cohesive and nuanced understanding of dietary practices in LIA Pada communities. Our integrated SIA (incl. modelling), ORA and microfossil records together with available zooarchaeological information show that Pada LIA community had a complex agriculturally oriented economy, with most likely C3 cereals forming their daily staple. Plant-based foods were supplemented by livestock primary (meat) and secondary (dairy) products in a range of 15–20%, with domination of sheep/goat/cattle over pig. The latter is evidenced by our SIA models as well as zooarchaeological records. Aquatic fauna, covering both mammals and fish, seem to have formed a rather small proportion of consumed food sources. Similar tendencies in historical peasant foodways—*i.e.* reliance on cereal-based foods, supplemented seasonally by smaller proportions of terrestrial animal proteins or aquatic sources (depending on the location)—have been recorded in local ethnographic sources as well (Moora, 2007).

On a societal level, the Pada community stands out as highly egalitarian, at least when comparing access to different food sources. There are no or only minimal dietary differences in SIA-based lifetime dietary records across men, women, and children but also across different age groups (Figs. 2 and 3). Even the children seem to have had diets like adults, and no socially expected patterns like children's food resembling the diets of women were detected. Notably, there is no food source distinction in grave good pottery either – funerary ritual meals concealed with the deceased do not vary between sexes, or adults and children, but are scattered across different food groups. This distinguishes the Pada community from several other (eastern) Baltic Late Iron Age and Early Medieval contexts, where previous studies have shown differences in lifetime and ritual diets by sex (Morrone *et al*., 2023; Oras *et al*., 2018; Pētersone-Gordina *et al*., 2022), provenance (Aguraiuja-Lätti & Lõugas, 2019), or social status (Linderholm *et al*., 2008; Malve *et al*., 2023), although few examples of non-stratified dietary practices have been noted as well (Danielisová *et al*., 2025; Etu-Sihvola *et al*., 2022; Pētersone-Gordina *et al*., 2023).

The only clearer social differentiation emerging from Pada material is the distinction of daily and burial ritual-related dietary practices. Funerary ritual vessels are dominated by terrestrial and aquatic animals (although note the potential ORA extraction bias above), whilst habitation site vessels have more indicators of plant-based food. The latter correlates well with the microfossil and SIA records discussed above, showing that daily foodways were dominated by plants. However, the tendency towards having more protein-rich and animal-based foods in ritual context has been also previously noted across the Baltics (Isaksson, 2000, 2003; Oras *et al*., 2018), so this trend seems to have some wider cultural significance. The second subtle difference between the two contexts is that the indicators of millet (although at trace level) were found in grave good pottery only. However, the significance of this finding is currently difficult to interpret, although the small number of millet grains in local archaeobotanical records (Johanson *et al*., 2024; Sillasoo & Hiie, 2007) might indicate a certain special value of millet as a cereal type. Although subtle, these deviations between the contents of habitation site and ritual pottery exemplify the importance of conducting comparative analysis of multiple and contemporaneous

archaeological contexts. This is the only way to create a more cohesive understanding of the overall dietary practices and the deviations of foodways across different socially embedded consumption contexts (Hastorf, 2017; Twiss, 2019).

Conclusions

We have presented a detailed multilevel analytical approach studying and combining a wide selection of dietary proxies: skeletal remains including bone, teeth and dental calculus, pottery, and zooarchaeological records. Our findings exemplify the major challenges of reconstructing past foodways through biomolecular and microarchaeological records, including analytical biases of single methods and/or proxies that can lead to contradictory results or even misinterpretation of dietary records.

By applying this combined approach, integrating different dietary datasets, analytes, and archaeological contexts, we have built a more coherent understanding of major food sources consumed in LIA Eastern Baltic. We have proven that the main staple of these communities was most likely C3 plants (cereal-based) foods, with terrestrial and aquatic substances providing smaller contributions to daily dietary intake. On the contrary, higher-trophic level foods, mostly consisting of terrestrial and aquatic animal-based products, were preferred in the (burial) ritual contexts. No dietary variations emerged across different sex and age groups either in lifetime or ritual (grave good pottery) contexts, meaning that local foodways in the Pada community were rather homogeneous and there was no food-based dietary stratification. All in all, our study highlights that a special effort must be made to collect and compare datasets from (a) different proxies, (b) different methods, and (c) multiple contexts to obtain a more cohesive understanding of ancient dietary practices.

However, the study could be further extended by adding similar comparative datasets from other spatiotemporal contexts, as some other regional and temporal deviations in the patterns outlined here might emerge. Also, currently our comparative approach was rather generic, but in the future a more systematic, perhaps also statistically advanced approach for combining and comparing the datasets from various dietary proxies ought to be developed. Here the recent advancements of statistical modelling and machine learning (Crabtree *et al*., 2017, 2021; Fernée & Trimmis, 2021) might be of value to increase the scientific rigour of multiproxy and multianalyte dietary analysis on a larger scale.

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Author Contribution E.O. worked out the article conceptualisation. E.O., M.T., E.R., K.J., A.U. and S.C. wrote the main manuscript text, compiled the figures and tables, and relevant SI materials. E.O., M.T.,

E.R., K.J., A.U., S.C., A. deV., H.S., A.M., S.I. collected the data and conducted formal analysis. All authors reviewed the manuscript.

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Data Availability All the data is provided in the article and the electronic supplementary information materials.

Declarations

Competing Interests The authors declare no competing interests.

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